

Physical Materials

Physical Materials are used to define the response of a certain body when it comes in contact with a different body during the gameplay. They are fairly easy to use and creating one physical material will give you certain default values that are identical to the physical values of the same material in the real world so it gives your game a realistic touch when the players are going through the levels.

Property	Description
Friction	This describes the Friction value of the surface of the material which decides how easily an object will slide when it makes surface contact with another object of same or different material.
Friction Combine Mode	This property enables the programmer to define how Friction Properties of materials are combined. While it is set at Average by default, one can override it using the Override Friction Combine Mode.
Restitution	This is basically the elasticity or how “bouncy” or how much energy a particular physical object will retain when it collides with another object of the same or other material complexions.
Restitution Combine Mode	This property enables the programmer to define how Restitution Properties of materials are combined. While it is set at Average by default, one can override it using the Override Restitution Combine Mode.
Density	This property is used with a Physical Object to calculate its mass properties. The higher the density, the heavier the object is. It is defined in gram per centimetre cube.
Surface Types	Surface Types of various objects used throughout the game are set in DefaultEngine.ini file dedicated for your project. It is responsible to define an enumeration that can be used in order to define multiple things from what sound a character makes when it walks to the type of damage an explosion should leave on a particular surface and so on and so forth.
Raise Mass to Power	This property is used to define the ratio in which the mass of an object increases as its size is increased. This is then applied to the mass as calculated on the basis of a solid object. In actuality as solid objects in the unreal world become bigger, they are more likely to be like a shell rather than a full solid of sorts.

Physics Bodies

As the complexity of 3D meshes in ever-increasing, proxies are mostly used to drive physical simulations in real-time 3D rendering applications. Physical Bodies or BodyInstances are simplified Physical 3D Meshes that the Unreal engine uses for real-time physical simulation any given point in time.